

JAVA SYLLABUS

Part 1 : Fundamental:

- Introduction to Java
- Pre Programming
- History and Introduction to Java
- Structure of Java Program
- Tokens of Java
- Data Types and Variables
- Operators
- Typecasting
- Decision Making Statements
- Loops
- Methods
- Dynamic Reading – Scanner Class
- Array

Part 2 : Object Oriented Programming

- Static Members
- Non-Static Members
- Object in Java
- Object Oriented Programming Introduction
- Constructor Overloading
- Constructor Chaining
- Principles of OOPS
- Encapsulation
- Inheritance – Types
- Polymorphism – Types
- Abstraction
- Non-Primitive Typecasting
 - Upcasting
 - Downcasting
- Singleton Class

Part 3 :

- Object Class and Methods
- String Class, Features and Methods
- Difference Between String, StringBuffer, StringBuilder
- Exception Handling
- Wrapper Classes
- Array of Objects
- Collection Frameworks
 - Collection Hierarchy
 - Map Hierarchy
- File Handling

Data Structure

- Loop
- Number and Pattern programs
- Time and Space Complexity
- Array
- String
- Linked List
- Stack
- Queue
- Tree
- Graph
- Dynamic Programming
- All Sorting Algorithms
- All Searching Algorithms
- LeetCode 400+ questions

WEB TECH SYLLABUS

1. Introduction
2. Doctype
3. Tag
4. Elements
5. Attributes
6. Comments
7. Text Formatting Tags
8. Heading Tags
9. Paragraph Tag
10. Image Tag
11. Hyperlink
12. List
13. Table
14. Form
15. Iframe
16. Video Tag
17. Audio Tag
18. Div And Span
19. Class And Id
20. Details And Summary Tag
21. Semantic Tags
22. Html5 Attributes
23. Html5 Assignment

CSS Syllabus

1. Introduction to CSS
2. History
3. Types of CSS
4. what is Selectors
5. Simple Selector
6. Combinator Selector
7. Attribute Selector
8. Pseudo Class Selector
9. Pseudo Element Selector
10. what is Property
11. Color Property
12. Background Property
13. Font Property
14. Text Property
15. Vendor Prefix
16. CSS Box Model Rule
17. Box Sizing Property
18. Overflow Property
19. Box shadow Property
20. Margin Property
21. Padding Property
22. Border & Border radius Property
23. Display Property
24. Position Property
25. Transform Property
26. Animation Property
27. CSS Units
28. Responsive Designing using Media query.

1. History
2. Introduction
3. Javascript Runtime Environment
4. Token
5. Scope
6. Global Execution Context
7. Var Let Const
8. Function
9. Practice Questions
10. Lexical Scope
11. Closure
12. Interview Questions
13. Array
14. Object
15. Json
16. Call Apply Bind
17. Constructor Function
18. This Keyword
19. Destructuring
20. Rest And Spread
21. Prototype
22. String
23. Dom
24. Event
25. Exception Handling
26. Non-Promise based APIs
27. Promise
28. Fetch API
29. Project-1
30. Project-2



React JS Syllabus

React.js Syllabus (by Sarvesh Sir)

1. Introduction to React

- What is React?
- Multi Page Applications (MPA) vs Single Page Applications (SPA)
- Advantages & Disadvantages of SPA
- History of React.js
- Why React was introduced

2. React Fundamentals

- Features of React.js
- Declarative Approach & Virtual DOM
 - Diffing Algorithm
 - Reconciliation & Patching
- Importance of key prop

3. Components

- Concept of Components
- Types of Components
 - Class-based Components (Stateful)
 - Function-based Components (Stateless)
- Difference between Class & Functional Components
- Rendering Components

4. JSX (JavaScript XML)

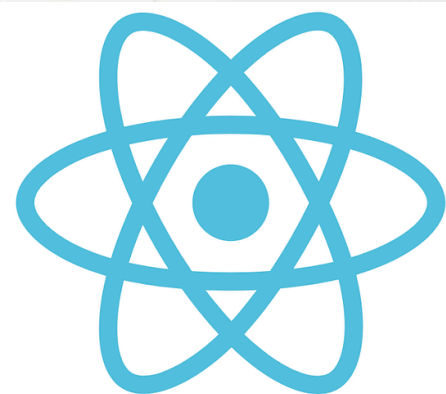
- Introduction & Syntax
- Rules of JSX
 - Parent wrapper / Fragments
 - Self-closing tags
 - Using Expressions in JSX
 - CamelCase conventions (className, htmlFor, events)

5. State and Props

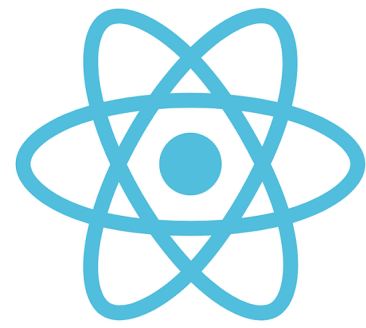
- State in Class vs Function Components
- Introduction to Hooks
 - useState() hook
- Props & Props Children
- Props Destructuring
- Difference between State and Props
- Props Drilling & Problems

6. Context API & State Management

- Introduction to Context API
- createContext, Provider, Consumer
- Using useContext() hook
- useReducer() hook
- Higher-Order Components (HOC)



React JS



7. Styling in React

- Ways of Adding CSS in React
 - Global CSS
 - CSS Modules
 - Conditional Rendering
-

8. Refs & Forms

- Introduction to Refs (useRef)
 - Controlled Forms
 - Uncontrolled Forms
-

9. Lifecycle Methods

- Component Lifecycle in Class-based Components
 - Mounting, Updating, Unmounting
 - Lifecycle methods: constructor, render, componentDidMount, componentDidUpdate, componentWillUnmount
 - Equivalent lifecycle using useEffect hook
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10. JavaScript Essentials for React

- Promises (Pending, Resolved, Rejected)
 - Handling with .then/.catch
 - Handling with async/await
 - Array methods (map, etc.)
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11. Performance Optimization

- React Memo (React.memo)
 - useMemo() hook
 - useCallback() hook
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12. Routing in React

- Introduction to React Router
 - BrowserRouter, Routes, Route
 - Nested & Shared Routing (<Outlet/>, index props)
 - Link vs NavLink (Differences)
 - Error Page handling
 - Navigation with useParams, useNavigate
 - createBrowserRouter & RouterProvider
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13. API Integration

- Axios Library
 - Installation & Setup
 - Fetching data with async/await
 - Fetching data with .then/.catch
- Displaying API Data

Modern Advanced Java Syllabus

(2025 Edition)

JEE (Jakarta Enterprise Edition)

1. JDBC-API to connect and communicate from java app to DB
2. Servlet-API and JSP for java web application

Frameworks:

1. Hibernate and JPA

2. Spring framework

1. spring-core
2. spring-boot
3. spring-data-jpa
4. spring-actuator
5. spring mvc

Microservices:

1. Monolithic vs Microservices
2. Basics of spring cloud

SQL syllabus

- Data / Database
- CRUD Operations
- DBMS
- RDBMS
- Relational Model
- E. F. Codd Rules
- Data Types
- Constraints
- Overview of SQL Statements
- SQL Categories:
 - DDL (Data Definition Language)
 - DML (Data Manipulation Language)
 - TCL (Transaction Control Language)
 - DCL (Data Control Language)
 - DQL (Data Query Language)
- SQL Concepts:
 - Operators
 - Functions
 - types of Functions
 - GROUP BY
 - HAVING
 - Subquery
 - Case 1
 - Case 2
 - MAX & MIN
 - Types of Subqueries
 - Nested Subquery
 - Employee–Manager Relationship
 - Joins
 - CROSS JOIN
 - INNER JOIN
 - OUTER JOIN
 - SELF JOIN
 - NATURAL JOIN
 - Co-related Subquery
 - Single Row Functions
 - Normalization

GEN AI syllabus

Module 1 : Introduction to Git

- What is a Version Control System
- Installation of Git and Tools
- Command and Concepts
- Working with Branches
- Handling Different Conflicts

Module 2 : Introduction to GitHub

- What is GitHub and how does it work?
- Understanding of Remote Repositories, Local Repository
- Connecting local repo to remote repo
- Push, Pull, clone, Fetch, merge
- Understanding SSH vs HTTPS authentication

Module 3 : Collaboration and Team Work

- Forking a repository
- Forking a repository
- Creating a Pull Request (PR)
- Code review and merging PRs
- Understanding upstream and origin remotes
- Resolving conflicts in team collaboration
- GitHub Flow and Feature Branch Workflow

Module 4 : Undoing & Reverting Changes

- Restore files
- Move commits back
- Undo commit safely
- Discard changes / Detached Head
- Remove untracked files
- Difference between reset, revert, restore
- Cheery Picking

Module 5 : Introduction to GitHub Desktop

- What is GitHub Desktop
- Difference between Git CLI and GitHub Desktop
- Why developers use GitHub Desktop (advantages)
- Installing GitHub Desktop (Windows/Mac)
- First-time setup (login with GitHub, configuring name & email)

Module 6 : Introduction to AI/ML Basics

- What is Artificial Intelligence
- Real-life examples of AI
- Difference between AI and Machine Learning
- What is Machine Learning
- How machines learn from data
- Types of Machine Learning (Supervised, Unsupervised)
- What is a Prompt and Prompt Engineering
- How AI understands prompts
- How to write good prompts
- Real-life uses of AI tools (ChatGPT, Copilot, Cursor)

GEN AI syllabus

Module 7 : Introduction to Github Copilot

- Installation & setup
- VS Code integration
- Enabling/Disabling Copilot
- Inline code suggestions
- Comment-based code generation
- Multiple suggestion cycling
- Copilot Chat usage
- Code completion & refactoring
- Working with frameworks (React, Node.js, etc.)
- Writing tests & docs with Copilot
- Shortcuts & settings
- Best practices & limitations
- Real-world use cases

Module 8: Introduction to Cursor AI

- Installation & setup
- Cursor AI vs VS Code comparison
- AI-powered autocomplete
- Inline code generation
- AI chat & code explanation
- Comment-based code writing
- Debugging with AI suggestions
- File-level & project-level context
- Code refactoring & optimization
- Command palette AI tools
- Integration with Git & GitHub
- Custom AI prompts in editor
- Best practices & limitations

Module 9 : Project I

- Todo App
- Layout Design

Module 10: Project II

- Food Delivery App (Zomato, Swiggy, Blinkit, Zepto)
- Education and Learning Website